

# Knowledge Engineering Process Report

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A Wi-Fi Troubleshooting Expert System

## **INTRODUCTION**

Expert systems are a key application of Artificial Intelligence that attempt to replicate the decision-making abilities of human experts within a specific domain. In knowledge engineering, expert knowledge is captured, structured, and represented in a way that allows a computer system to perform reasoning and provide guidance similar to that of a domain specialist.

This document presents a simple rule-based diagnostic expert system designed to identify common causes of Wi-Fi connectivity problems on a laptop. The system operates by asking the user a series of structured diagnostic questions regarding the laptop's network status and connectivity behaviour. Based on the responses provided, the system evaluates a predefined set of conditions and determines the most likely cause of the problem.

Each potential diagnosis in the system is associated with a confidence score. These scores represent the level of belief that a particular condition is responsible for the observed symptoms, based on the judgement and experience of domain experts. The system then provides explanations and troubleshooting advice to help resolve the identified issue.

## **PROBLEM DEFINITION**

The project focuses on diagnosing common laptop Wi-Fi connectivity problems. These include situations where the Wi-Fi network is not visible, the device cannot connect to the network, or the device connects but the internet does not work. The system attempts to identify the most likely cause of the problem based on user responses.

## FINAL DIAGNOSIS AND CONFIDENCE SCORES

The names assigned to conditions remain constant throughout the diagnosis process to maintain consistency within the knowledge base. Each condition is associated with a confidence score that represents the degree of certainty about the relationship between the observed symptoms and the possible diagnosis. In this system, the confidence scores are assigned based on the judgement and experience of domain experts, who estimate how strongly certain symptoms indicate certain conditions. These values serve as a measure of belief in the rules defined within the system. Although the scores are initially determined through expert knowledge rather than strict mathematical computation, they may be reviewed and adjusted over time as new knowledge about the domain becomes available or as improvements are made to the system.

|           | <b>CONDITIONS FOR DIAGNOSIS</b> | <b>CONFIDENCE<br/>SCORE</b> |
|-----------|---------------------------------|-----------------------------|
| <b>1.</b> | airplane_mode_enabled           | <b>95</b>                   |
| <b>2.</b> | wifi_disabled                   | <b>90</b>                   |
| <b>3.</b> | adapter_or_driver_issue         | <b>80</b>                   |
| <b>4.</b> | password_or_profile_issue       | <b>85</b>                   |
| <b>5.</b> | dns_or_ip_stack_issue           | <b>80</b>                   |
| <b>6.</b> | router_or_isp_problem           | <b>75</b>                   |

## SYMPTOMS / FACTS (8 QUESTIONS MAX)

Accept Yes or No answers only

| Fact name              | Question (asked to user)                      |
|------------------------|-----------------------------------------------|
| airplane_mode          | Is Airplane Mode ON?                          |
| wifi_enabled           | Is Wi-Fi enabled on the laptop?               |
| sees_networks          | Can the laptop see Wi-Fi networks?            |
| can_connect            | Can it connect to the Wi-Fi network?          |
| connected_to_wifi      | Does it show connected to Wi-Fi?              |
| internet_access        | Does the internet work on the laptop?         |
| other_devices_connect  | Do other devices connect to that same Wi-Fi?  |
| other_devices_internet | Do other devices have internet on that Wi-Fi? |

## KNOWLEDGE ACQUISITION

Knowledge about Wi-Fi troubleshooting was gathered from networking troubleshooting guides and technical resources. These sources explain common network problems and the steps technicians normally follow when diagnosing them.

Sources used during knowledge acquisition included:

- Cisco Networking Basics troubleshooting documentation
- How-To Geek articles on common Wi-Fi connection problems
- GeeksforGeeks explanations of expert systems and rule-based reasoning
- General IT troubleshooting practices used in laptop support

From these sources, six common diagnoses were identified for the system:

airplane\_mode\_enabled, wifi\_disabled, adapter\_or\_driver\_issue, password\_or\_profile\_issue, dns\_or\_ip\_stack\_issue, and router\_or\_isp\_problem.

## DIAGNOSIS CONDITIONS (REQUIRED + NEGATIVE CONDITIONS)

The diagnostic conditions have been arranged in an order that reflects their relevance to the diagnosis process. However, this ordering does not imply that a condition with a higher confidence score will always result in a correct diagnosis. Instead, the confidence scores represent the likelihood or degree of belief associated with each condition, indicating how strongly the available evidence supports a particular diagnosis.

|    | Diagnosis                 | Conditions                                                                                                                              |
|----|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1. | airplane_mode_enabled     | ✓ airplane_mode = yes                                                                                                                   |
| 2. | wifi_disabled             | ✓ airplane_mode = no<br>(negative condition)<br>✓ wifi_enabled = no                                                                     |
| 3. | adapter_or_driver_issue   | ✓ airplane_mode = no(negative)<br>✓ wifi_enabled = yes<br>✓ sees_networks = no                                                          |
| 4. | password_or_profile_issue | ✓ airplane_mode = no (negative)<br>✓ wifi_enabled = yes<br>✓ sees_networks = yes<br>✓ can_connect = no<br>✓ other_devices_connect = yes |
| 5. | dns_or_ip_stack_issue     | ✓ connected_to_wifi = yes<br>✓ internet_access = no<br>✓ other_devices_internet = yes                                                   |

|    |                       |                                                                                                                                                                                                                                                 |
|----|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6. | router_or_isp_problem | <p><b>Pattern A</b></p> <p>✓ other_devices_connect = no</p> <p><b>OR</b></p> <p><b>Pattern B</b></p> <p>✓ other_devices_internet = no</p> <p><i>(Programmers can implement this as an OR rule because one of these patterns is enough.)</i></p> |
|----|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## ADVICE (3–4 STEPS EACH)

### 1. airplane\_mode\_enabled

- Turn off Airplane Mode.
- Enable Wi-Fi afterwards if it remains off.
- Reconnect to the Wi-Fi network.
- Restart the laptop if Wi-Fi still doesn't appear.

### 2. wifi\_disabled

- Turn on Wi-Fi from settings (or the Wi-Fi key/switch).
- Confirm the Wi-Fi adapter is enabled in network settings.
- Reconnect to the correct network.
- Restart the laptop if Wi-Fi won't enable.

### **3. adapter\_or\_driver\_issue**

- a) Restart the laptop.
- b) Check the Wi-Fi adapter is enabled (Device Manager / Network settings).
- c) Update or reinstall the Wi-Fi driver.
- d) Test with a mobile hotspot or USB Wi-Fi adapter to isolate hardware issues.

### **4. password\_or\_profile\_issue**

- a) “Forget” the Wi-Fi network and reconnect.
- b) Re-enter the password carefully (confirm it’s the latest one).
- c) Restart the router and try again.
- d) Try connecting to a mobile hotspot to confirm the laptop can connect elsewhere.

### **5. dns\_or\_ip\_stack\_issue**

- a) Disconnect and reconnect to Wi-Fi.
- b) Renew IP / DHCP (or run the network troubleshooter).
- c) Flush DNS and retry.
- d) Temporarily change DNS to a public DNS (e.g., 8.8.8.8 or 1.1.1.1) to test.

### **6. router\_or\_isp\_problem**

- a) Restart the router (power off ~10 seconds, then on).
- b) Check the router’s Internet/WAN indicator lights.
- c) Try another network (mobile hotspot) to confirm the laptop is not the issue.
- d) Contact the ISP if the internet remains down for all devices.

## INFERENCE PROCESS

The inference engine evaluates the rules in the knowledge base and determines the most likely diagnosis. The Python interface collects user responses and sends them to the Prolog engine, which evaluates the rules and selects the matching diagnosis explained as follows:

### EXPLANATION TEXT (1–2 SENTENCES EACH)

1. **airplane\_mode\_enabled:** Airplane Mode disables wireless radios, including Wi-Fi, so the laptop cannot connect until it is turned off.
2. **wifi\_disabled:** Wi-Fi is turned off on the laptop, preventing it from scanning and connecting to networks.
3. **adapter\_or\_driver\_issue:** Wi-Fi is enabled and Airplane Mode is off, but no networks are visible, indicating the adapter may be disabled, malfunctioning, or the driver is faulty.
4. **password\_or\_profile\_issue:** The laptop sees networks but cannot connect while other devices can, suggesting incorrect credentials or a corrupted saved Wi-Fi profile.
5. **dns\_or\_ip\_stack\_issue:** The laptop is connected to Wi-Fi but has no internet while other devices do, indicating a DNS or IP configuration problem on the laptop.

6. **router\_or\_isp\_problem:** If multiple devices cannot connect or cannot access the internet, the issue is likely with the router or the ISP rather than the laptop.

## TEST CASES

The system was tested using several WiFi troubleshooting scenarios. Different combinations of symptoms were used to verify that the system produces the correct diagnosis. These test cases ensured that the inference engine correctly maps symptoms to conclusions. We use these for quick validation. (Inputs not listed can be “don’t care”, but giving full inputs helps).

| TEST CASE ID               | TEST CASE                                                      | EXPECTED OUTCOME        |
|----------------------------|----------------------------------------------------------------|-------------------------|
| TC1 — Airplane Mode        | airplane_mode = yes                                            | airplane_mode_enabled   |
| TC2 — Wi-Fi Disabled       | airplane_mode = no<br>Wifi_enabled = no                        | wifi_disabled           |
| TC3 — Adapter/Driver Issue | airplane_mode = no<br>wifi_enabled = yes<br>sees_networks = no | adapter_or_driver_issue |

|                                                                                                   |                                                                                                                    |                              |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>TC4 — Password/Profile Issue</b>                                                               | airplane_mode = no<br>wifi_enabled = yes<br>sees_networks = yes<br>can_connect = no<br>other_devices_connect = yes | password_or_profile_issue    |
| <b>TC5 — DNS/IP Stack Issue</b>                                                                   | connected_to_wifi = yes<br>internet_access = no<br>other_devices_internet = yes                                    | dns_or_ip_stack_issue        |
| <b>TC6 — Router/ISP Problem (No one connects)</b>                                                 | other_devices_connect = no                                                                                         | router_or_isp_problem        |
| <i>TC7 — Router/ISP Problem (Connected but no internet for everyone)</i><br><br><i>(Optional)</i> | <i>other_devices_connect = yes</i><br><i>other_devices_internet = no</i>                                           | <i>router_or_isp_problem</i> |

## CONCLUSION

The knowledge engineering process allowed the team to convert common WiFi troubleshooting knowledge into a rule-based expert system. By representing the

troubleshooting logic as rules in Prolog and connecting it to a Python interface, the system can analyze user symptoms and provide appropriate advice.

## **REFERENCES**

Russell, S., & Norvig, P. Artificial Intelligence: A Modern Approach.

Cisco Networking Basics – Wi-Fi Troubleshooting Guide.

How-To Geek – Common Wi-Fi Problems and How to Fix Them.

GeeksforGeeks – Expert Systems in Artificial Intelligence.

University lecture materials on Knowledge-Based Systems.